RPractical

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Part A: Basic Operations

#A1

```
v <- seq(5, 50, by = 5)
print(length(v))
```

```
## [1] 10
```

```
print(sum(v))
```

```
## [1] 275
```

```
print(mean(v))
```

```
## [1] 27.5
```

#A2

```
w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
print(w)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
print(z)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

#A3

```
print(17 %% 5)
```

```
## [1] 2
```

```
print(17 %/% 5)
```

```
## [1] 3
```

```
print(2^10)
```

```
## [1] 1024
```

```
print(sqrt(144))
```

```
## [1] 12
```

```
print(round(pi, 3))
```

```
## [1] 3.142
```

#A4

```
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)
print(a + b)
```

```
## [1]  3  7 11 15
```

```
print(a * b)
```

```
## [1]  2 12 30 56
```

```
print(a > b)
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
print(sum(a * b))
```

```
## [1] 100
```

#A5

```
print(paste("Student", 1:3, sep = "-"))
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```
print(class(42))                # numeric
```

```
## [1] "numeric"
```

```
print(class(42L))               # integer
```

```
## [1] "integer"
```

```
print(class("42"))              # character
```

```
## [1] "character"
```

```
print(class(41i))               # complex
```

```
## [1] "complex"
```

```
print(class(TRUE))              # logical
```

```
## [1] "logical"
```

Part B: Data Frames and Conditional Statements

#B1

```
students <- data.frame(
  id         = 1:8,
  name       = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex        = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section    = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm    = c(78, 85, 92, 67, 74, 88, 95, 59),
  final      = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)
print(students)
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

#B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
print(dim(students))
```

```
## [1] 8 7
```

```
print(colnames(students))
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
print(summary(students))
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance   
##  Min.   :65.00   Min.   : 60.00  
##  1st Qu.:77.00   1st Qu.: 81.75  
##  Median :81.00   Median : 89.00  
##  Mean   :82.12   Mean   : 86.00  
##  3rd Qu.:91.50   3rd Qu.: 95.75  
##  Max.   :96.00   Max.   :100.00
```

#B3

```
students[["average"]] <- (students[["midterm"]] + students[["final"]]) / 2
print(students)
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       A      92    96        100    94.0
## 4  4  Dana   F       C      67    71         72    69.0
```

```
## 5  5  Elif   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7 Grace   F       A      95    93         98    94.0
## 8  8  Hiro   M       B      59    65         60    62.0
```

#B4

```
students[["passed"]] <- students[["average"]] >= 70 & students[["attendance"]] >= 75
print(students)
```

```
##   id  name sex section midterm final attendance average passed
## 1  1 Alice   F       A      78    82         95    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE
## 3  3  Chen   M       A      92    96        100    94.0   TRUE
## 4  4  Dana   F       C      67    71         72    69.0  FALSE
## 5  5  Elif   F       B      74    80         90    77.0   TRUE
## 6  6 Farid   M       C      88    91         85    89.5   TRUE
## 7  7 Grace   F       A      95    93         98    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE
```

#B5

```
students[["standing"]] <- ifelse(students[["average"]] >= 90, "Honors",
                          ifelse(students[["average"]] >= 75, "Good", "Needs Support"))
print(students)
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4  4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5  5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6  6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7  7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
```

#B6

```
classify <- function(score) {
  if (score >= 85) {
    "Distinction"
  } else if (score >= 70) {
    "Merit"
  } else if (score >= 50) {
    "Pass"
  } else {
    "Fail"
  }
}
print(sapply(c(90, 73, 55, 40), classify))
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

#C1

```r
new_students <- data.frame(
  id         = c(9, 10),
  name       = c("Ines", "Jamal"),
  sex        = c("F", "M"),
  section    = c("C", "A"),
  midterm    = c(81, 70),
  final      = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)
new_students[["average"]]  <- (new_students[["midterm"]] + new_students[["final"]]) / 2
new_students[["passed"]]   <- new_students[["average"]] >= 70 & new_students[["attendance"]] >= 75
new_students[["standing"]] <- ifelse(new_students[["average"]] >= 90, "Honors",
                              ifelse(new_students[["average"]] >= 75, "Good", "Needs Support"))

students_all <- rbind(students, new_students)
print(students_all)
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

#C2

```r
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
print(students_all)
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
##    quiz
## 1     8
## 2     7
## 3    10
## 4     6
## 5     7
## 6     9
## 7    10
```

```
## 8     5
## 9     8
## 10    6
```

```
print(class(students_all))
```

```
## [1] "data.frame"
```

#C3

```
m <- cbind(midterm = students_all[["midterm"]], final = students_all[["final"]])
m <- rbind(m, mean = colMeans(m, na.rm = TRUE))
print(m)
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

#C4

```
# rbind() on data frames requires matching column names. If the names differ,
# it stops with the error "names do not match previous names". tryCatch()
# lets us catch that error, print the message, and continue without crashing.
bad <- students
names(bad) <- "student_name"

tryCatch(rbind(students, bad),
         error = function(e) {
           message("rbind failed: ", conditionMessage(e))
           NULL
         })
```

```
## rbind failed: names do not match previous names
```

```
## NULL
```

Part D: Extracting and Subsetting

#D1

```
print(getElement(students_all, "name"))
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
print(students_all[["name"]])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
print(students_all[, "name"])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

#D2

```r
print(students_all[2:5, c("name", "midterm", "final")])
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

#D3

```r
print(students_all[students_all[["attendance"]] > 90, ])
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```

#D4

```r
print(subset(students_all,
             sex == "F" & section %in% c("A", "C") & final >= 75,
             select = c(name, section, final)))
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

#D5

```r
print(students_all[["name"]][which.max(students_all[["average"]])])
```

```
## [1] "Chen"
```

```r
print(students_all[["name"]][which.min(students_all[["average"]])])
```

```
## [1] "Hiro"
```

#D6

```r
sorted <- students_all[order(students_all[["average"]], decreasing = TRUE), ]
print(head(sorted, 3))
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

Part E: Basic Factors

#E1

```r
students_all[["section"]] <- factor(students_all[["section"]])
students_all[["sex"]]     <- factor(students_all[["sex"]])
print(levels(students_all[["section"]]))
```

```
## [1] "A" "B" "C"
```

```r
print(levels(students_all[["sex"]]))
```

```
## [1] "F" "M"
```

#E2

```r
print(table(students_all[["sex"]]))
```

```
##
## F M
## 5 5
```

```r
print(table(students_all[["section"]]))
```

```
##
## A B C
## 4 3 3
```

```r
print(table(students_all[["sex"]], students_all[["section"]]))
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

#E3

```r
print(as.integer(students_all[["sex"]]))
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```r
# The integers are the internal codes of the factor levels. Levels are sorted
# alphabetically, so F = 1 and M = 2.
```

#E4

```r
students_all[["sex_label"]] <- factor(students_all[["sex"]],
                                      levels = c("F", "M"),
                                      labels = c("Female", "Male"))
print(levels(students_all[["sex_label"]]))
```

```
## [1] "Female" "Male"
```

#E5

```r
print(summary(students_all[["sex_label"]]))
```

```
## Female   Male
##      5      5
```

```r
print(mean(students_all[["average"]][students_all[["sex_label"]] == "Female"]))
```

```
## [1] 80.5
```

```r
print(mean(students_all[["average"]][students_all[["sex_label"]] == "Male"]))
```

```
## [1] 79.3
```